

5 (a) (i) $A = 3.00$ B1
 $\omega = 2\pi / T = 2\pi / 0.785 = 8.00$ B1

$$x = A \cos(\omega t - B)$$

$$2.00 = 3.00 \cos(8.00(0) - B)$$

$$B = 0.841$$

B1

(ii) Maximum velocity = $\omega x_0 = (8.00)(3.00) = 24.0$ A1

$$\text{Maximum acceleration} = \omega^2 x_0 = (8.00)(3.00)^2 = 192$$

A1

(iii) $\frac{1}{2} m v_{\max}^2 = \frac{1}{2} m \omega^2 x_0^2$
 $= \frac{1}{2} (5.0 \times 10^{-6})(8.00^2)(3.00^2)$
 $= 1.44 \times 10^{-3} \text{ J}$

1 for correct maximum kinetic energy

B1

1 for correct shape (2 periods with peak slightly to the left of origin)

B1

(b) An oscillator with damping can vibrate at resonance with amplitude that remains constant with time. M1

This is because the energy supplied by driver is equal to energy lost due to dissipative forces. A1

6 (a) When a p.d. of 240 V is applied across it, then the power dissipated